INCH-POUND
MIL-DTL-24231/5E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/5D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTORS, SINGLE CABLE, TYPE III, THREE-, FOUR-, OR FIVE-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 18.

2. The gland nut shall be tightened until the packing tightens against the wires. This can be determined by moving the wires gently back and forth until the wire is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold the wires in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires.

3. Full or partial threads shall not extend more than 0.063 inch beyond the ¾-inch thread length.

4. The alternate method of installing the guide pin (see figure 3) eliminates the need for welding in the fabrication of a single cable connector body.

5. Retainer plate, packing, and retainer plate assembly for other special conductor arrangements (for example, LS2SWU-1 and MIL-DTL-16878/36 BJE0 wire combinations) shall meet the following requirements:

   a. Where LS2SWU-1 cable is required, the hole size shall be 17/64 diameter.
   b. Positional tolerancing for all holes shall be ⌖ | Ø.010 |.
   c. Holes shall not be located closer than 0.010 from the minimum gland nut inside diameter of part number M24231/5-072 and the retainer plate position pin holes of part number M24231/5-050.
   d. Minimum material between holes shall be 0.031.
   e. Protective sleeving shall be sized to suit the number of cables required where use of the specified sleeving, item 24, is inadequate.

6. The retainer plates and the retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with Class A, Form 1, hot-finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with Class A, Form 1, cold-drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.

7. Welding of the pin (/5-071) is allowed after the hull fitting body has been age-hardened to achieve its required physical properties.

8. Unless otherwise specified (see figure 17), all dimensions are in inches.

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9. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

10. The flange outside diameter is intended to coincide with commonly available 2¾-inch bar stock. The dimension may vary after machine clean-up.

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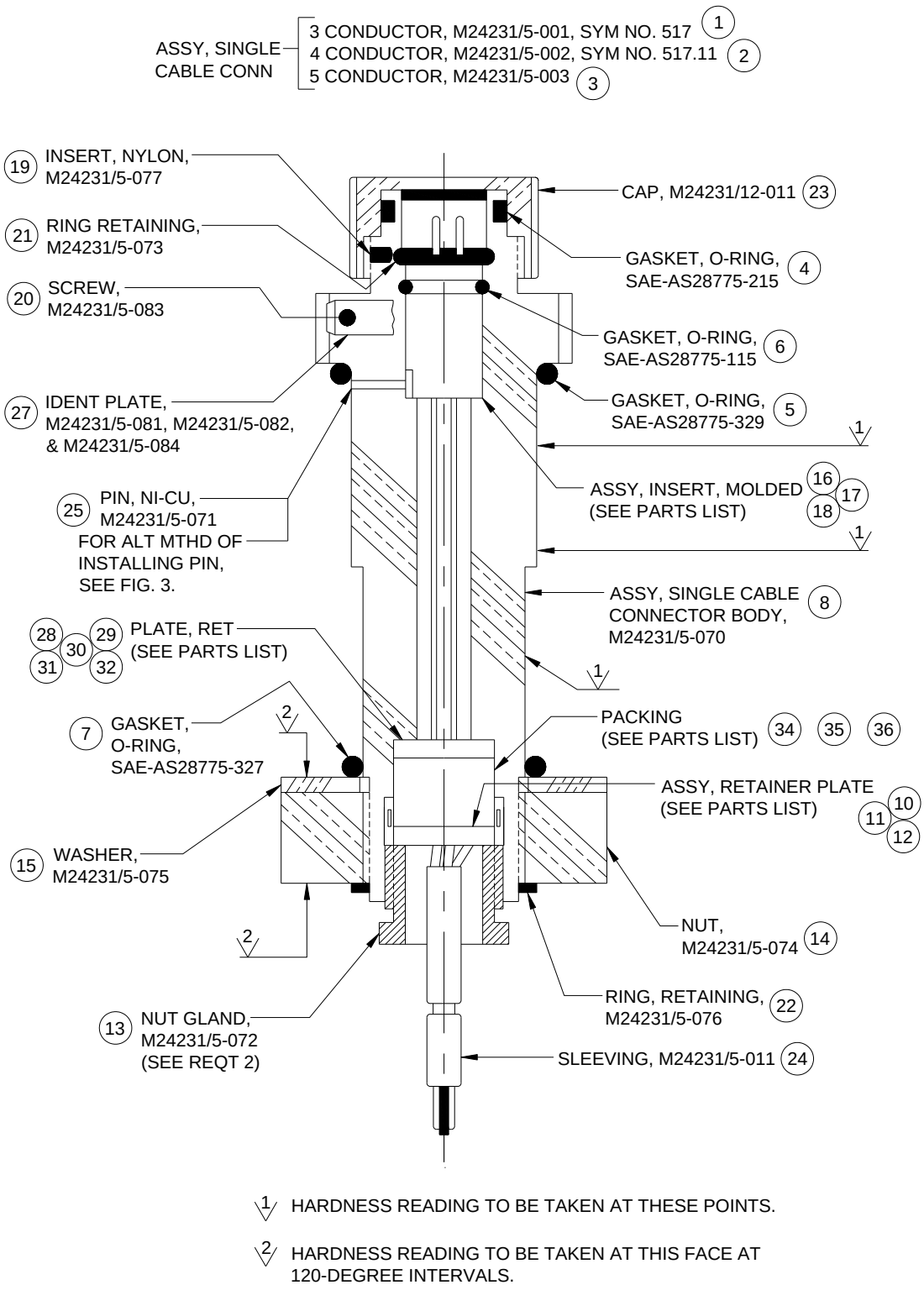


FIGURE 1. Connector, single cable – typical.

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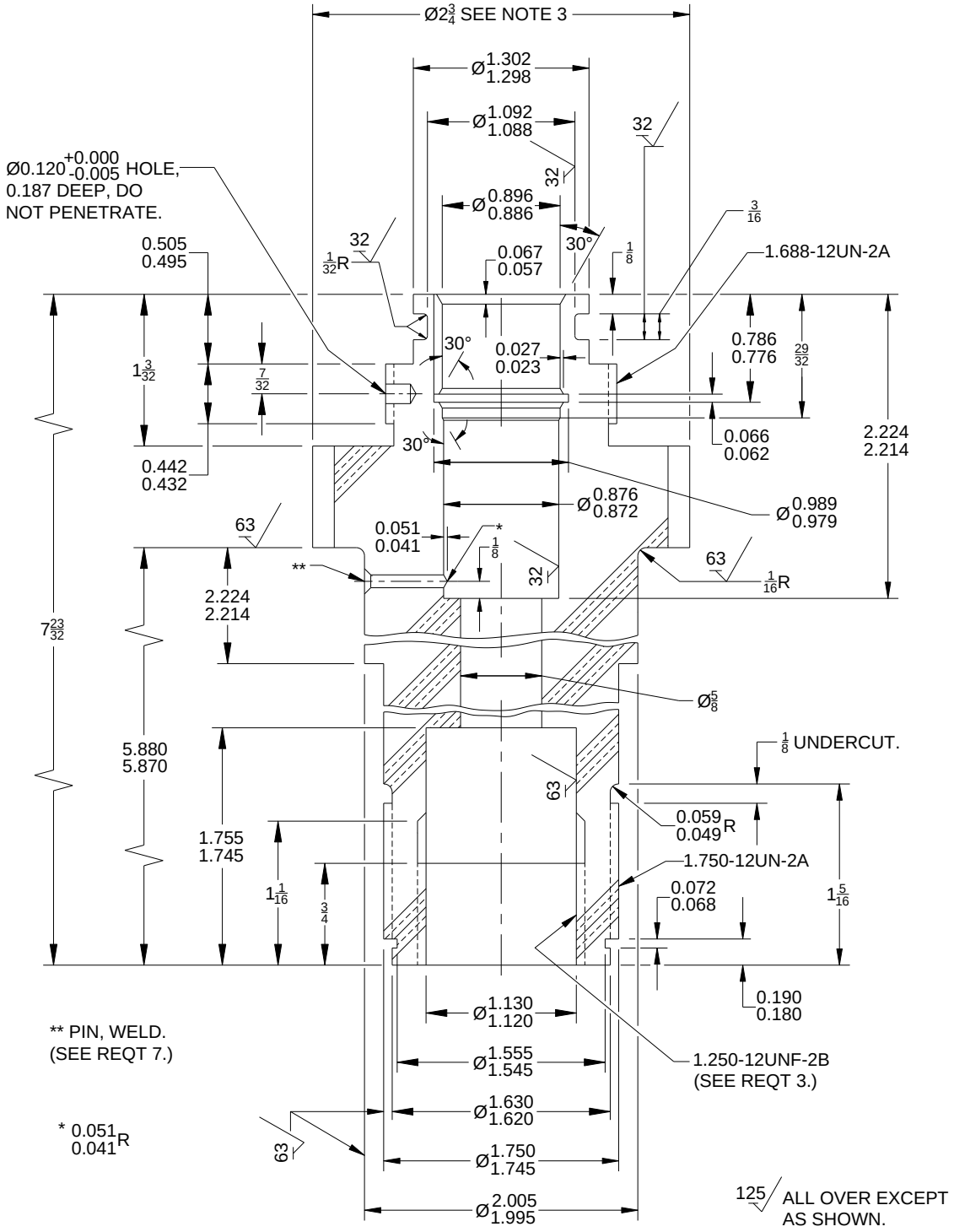


FIGURE 2. Assembly, single cable connector body – M24231/5-070. (8)

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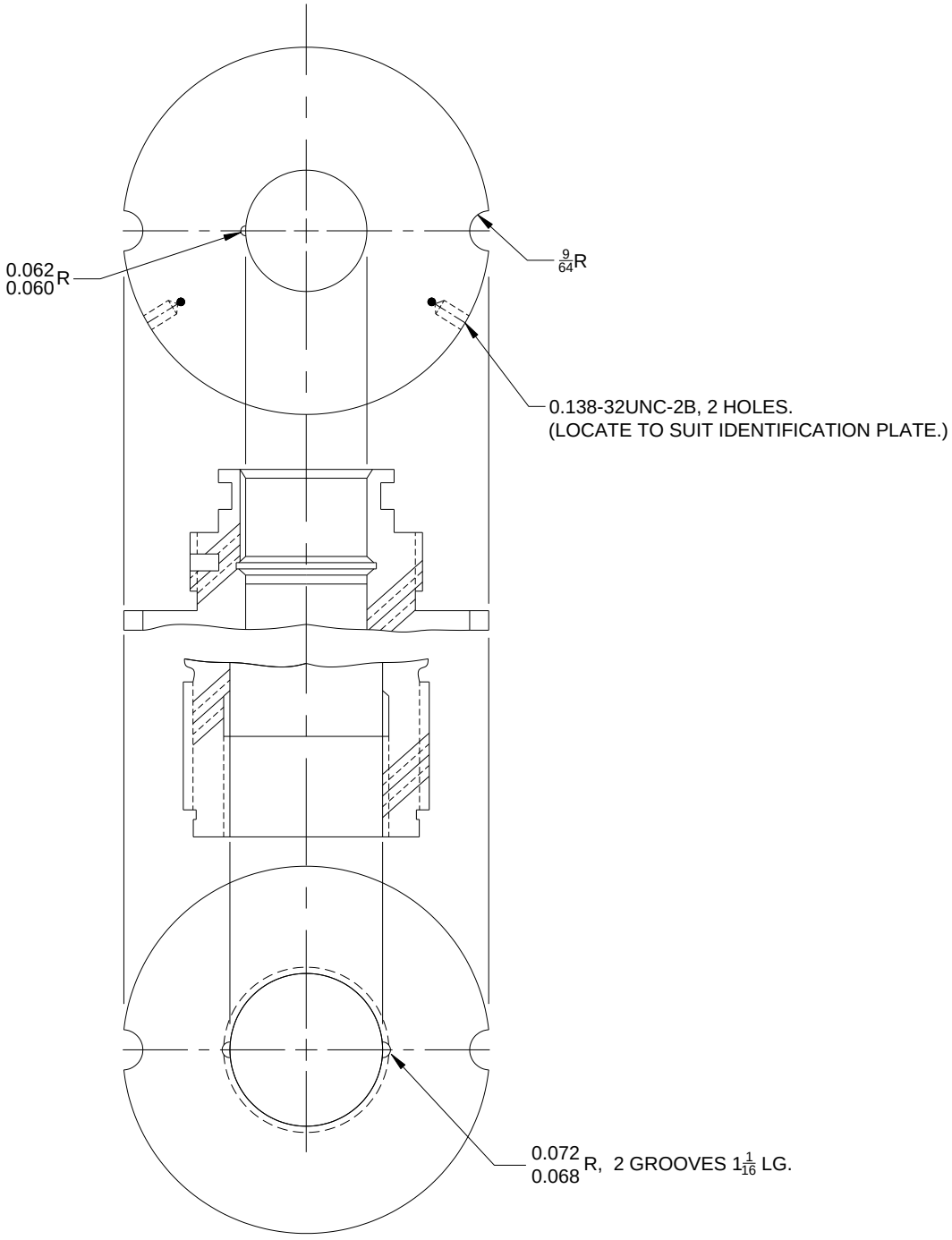
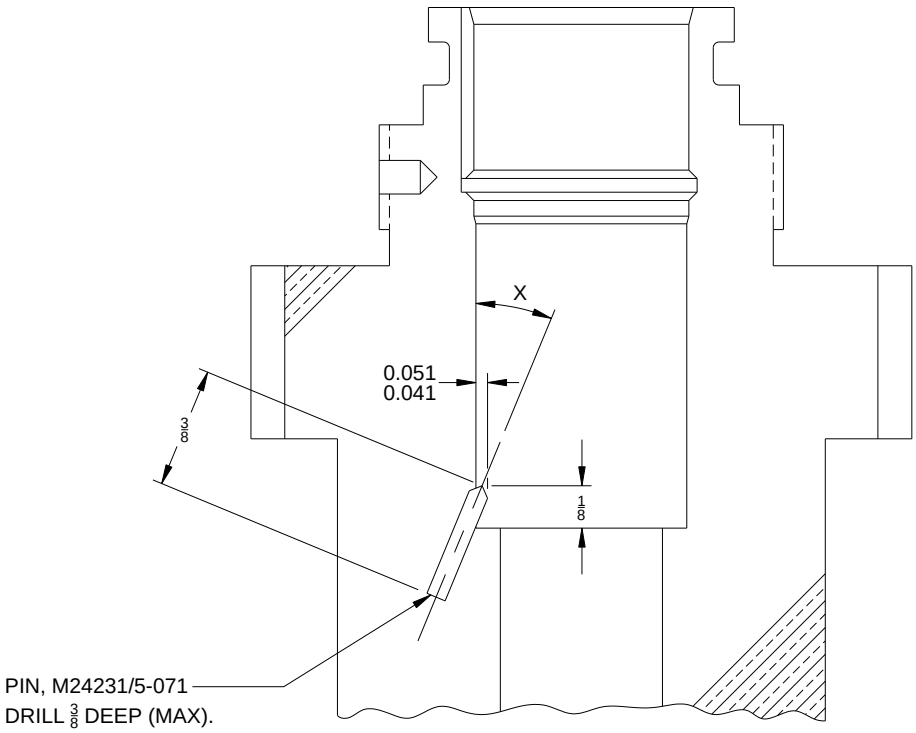


FIGURE 2. Assembly, single cable connector body – M24231/5-070 – Continued. (8)

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CONNECTOR BODY	"X" DEGREES
M24231/5-070	$21\frac{1}{2}$
M24231/6-010	$21\frac{1}{2}$
M24231/7-040	27
M24231/8-020	27
M24231/9-030	27
M24231/9-040	27
M24231/10-040	35
M24231/19-010	40

NOTE: INSTALL PIN WITH LOCKING AND RETAINING COMPOUND IN ACCORDANCE WITH GRADE "A" OF MIL-S-22473.

FIGURE 3. Alternate method for installation of guide pin (see requirement 4). (25)

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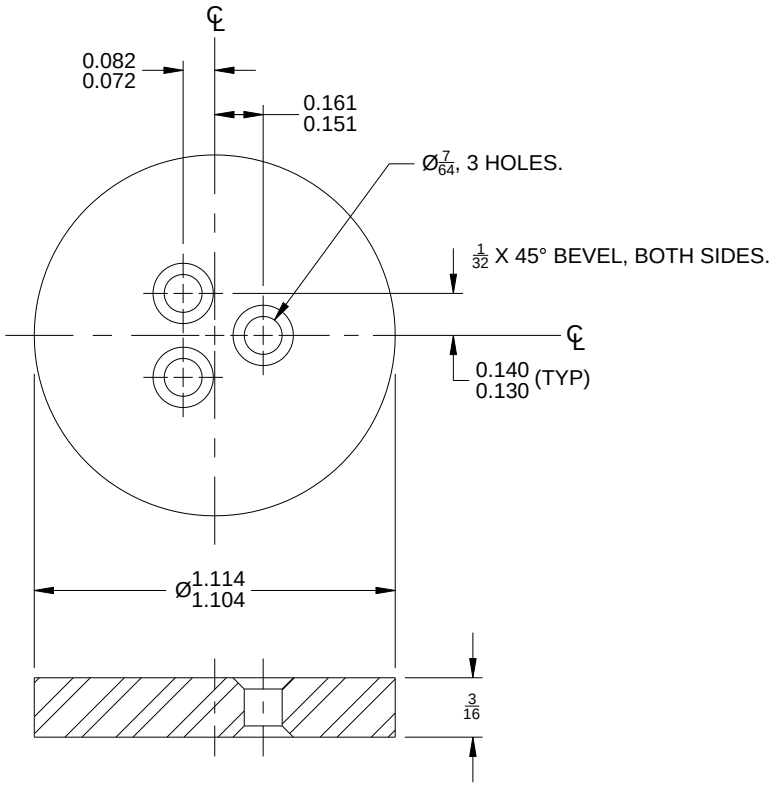


FIGURE 4. Plate, retainer M24231/5-051. (28)

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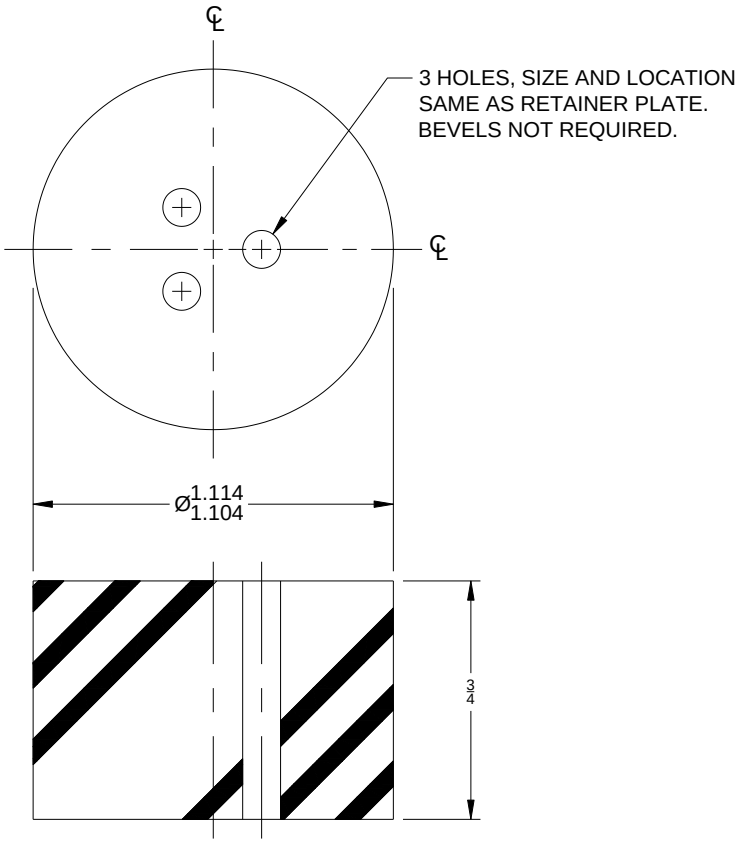


FIGURE 5. Packing – M24231/5-052. (34)

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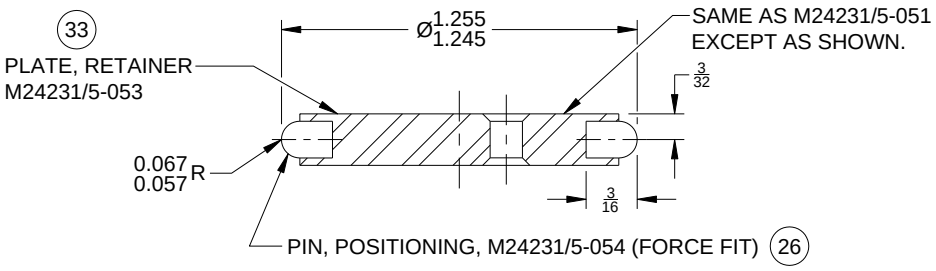


FIGURE 6. Assembly, retainer plate – M24231/5-050. 10

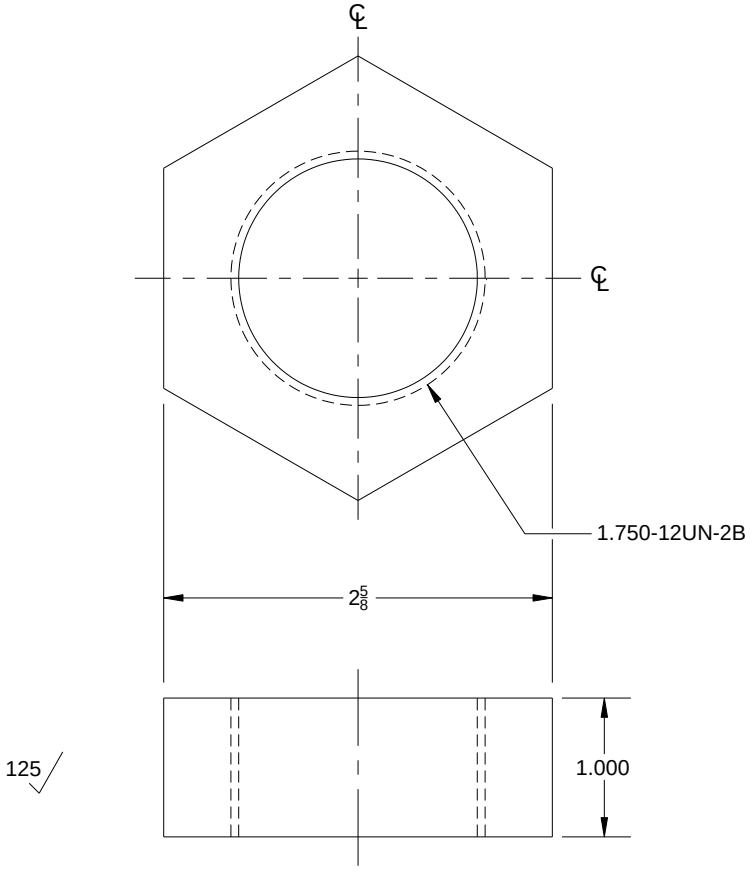


FIGURE 7. Nut – M24231/5-074. 14

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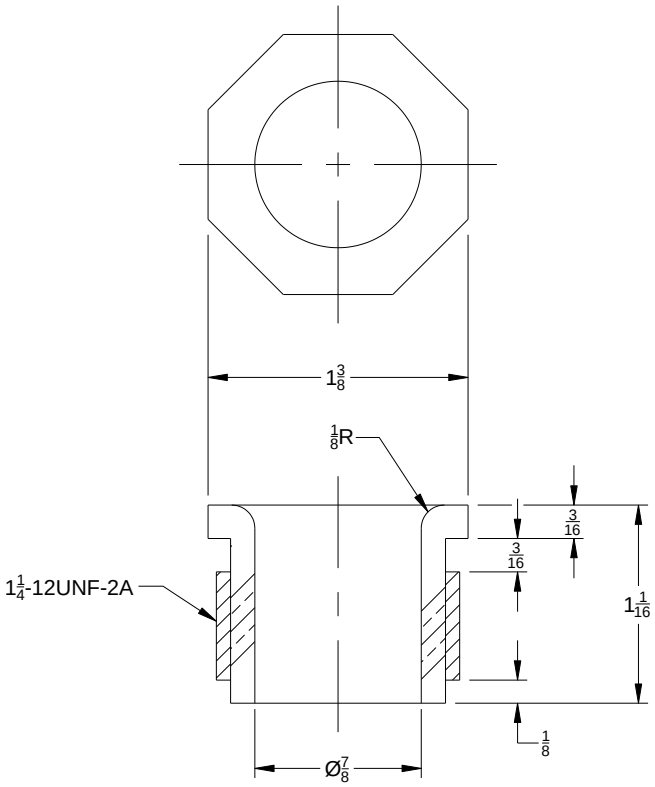


FIGURE 8. Nut, gland – M24231/5-072. (13)

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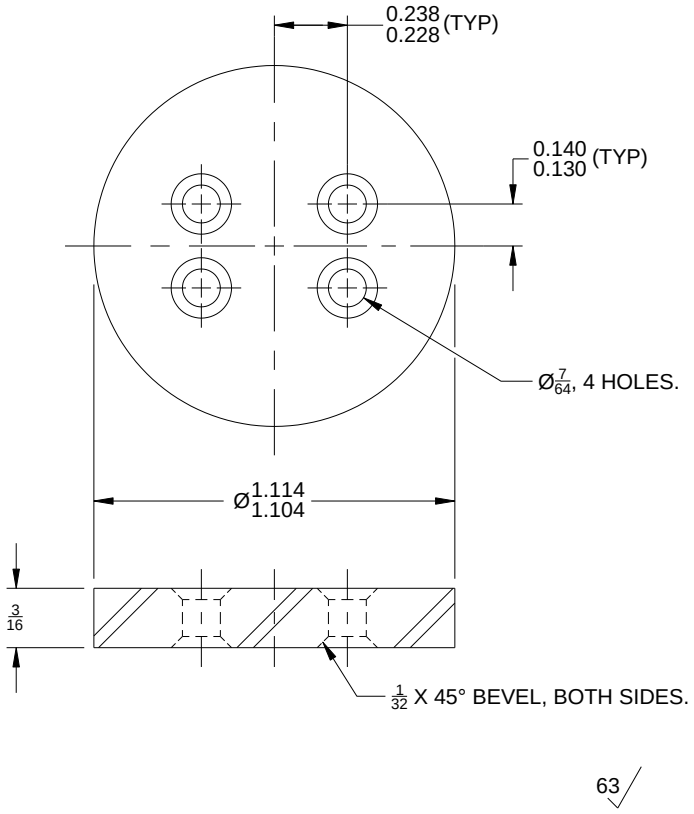


FIGURE 9. Plate, retainer – M24231/5-061. (29)

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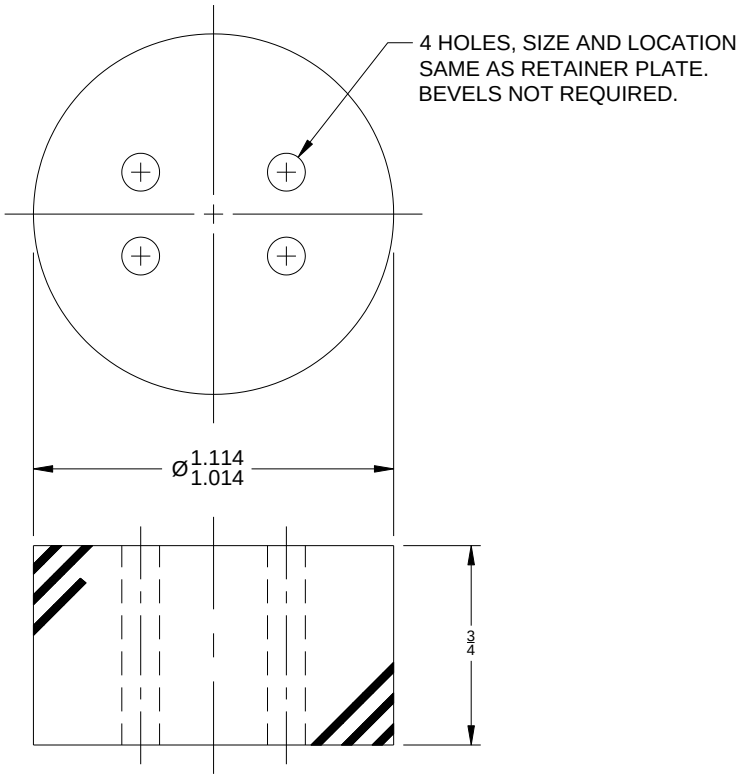


FIGURE 10. Packing – M24231/5-063. (35)

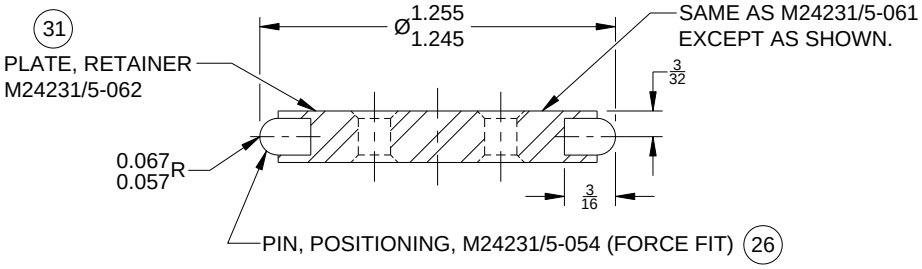
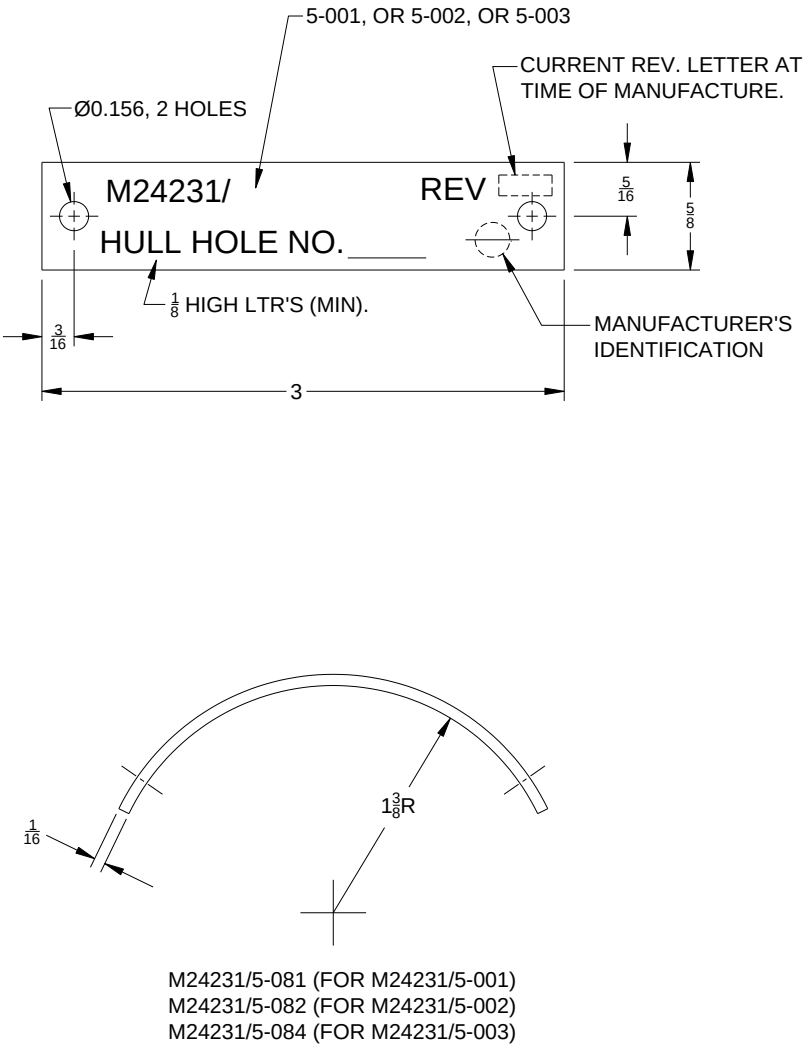


FIGURE 11. Assembly, retainer plate – M24231/5-060. (11)

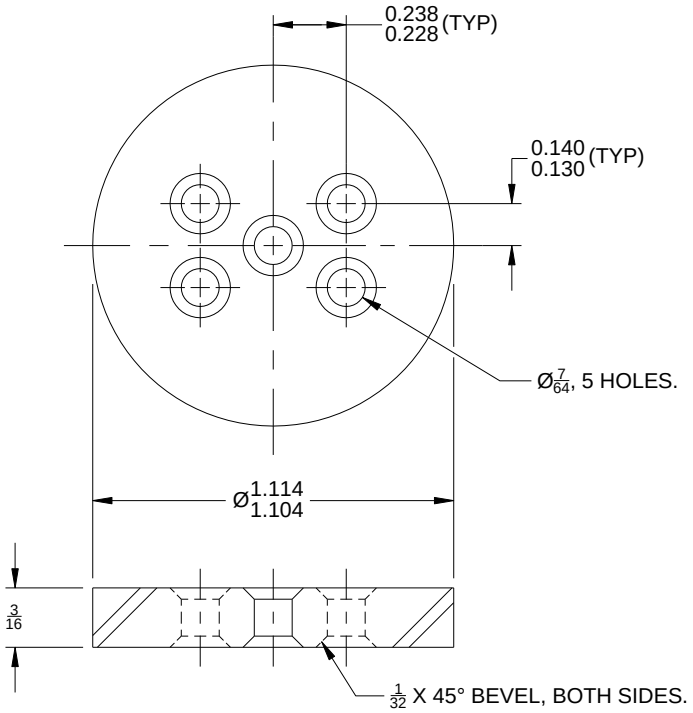
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M24231/5-081 (FOR M24231/5-001)
M24231/5-082 (FOR M24231/5-002)
M24231/5-084 (FOR M24231/5-003)

FIGURE 12. Plate, identification. (27)

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SEE REQUIREMENT 5.

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FIGURE 13. Plate, retainer – M24231/5-064. (30)

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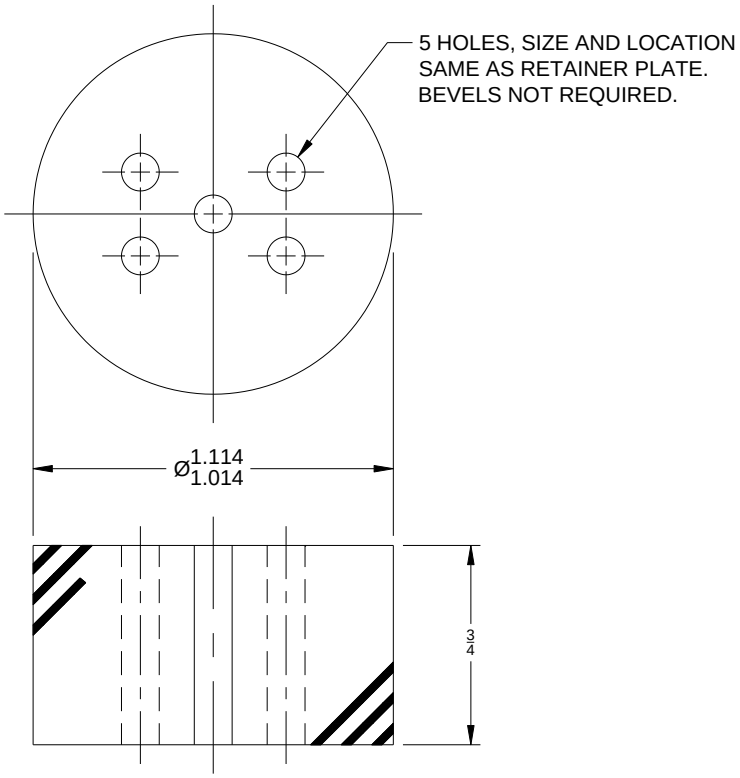


FIGURE 14. Packing – M24231/5-085. (36)

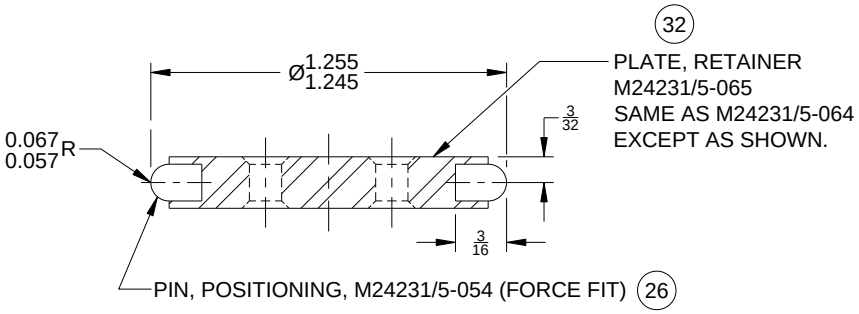


FIGURE 15. Assembly, retainer plate – M24231/5-066. (12)

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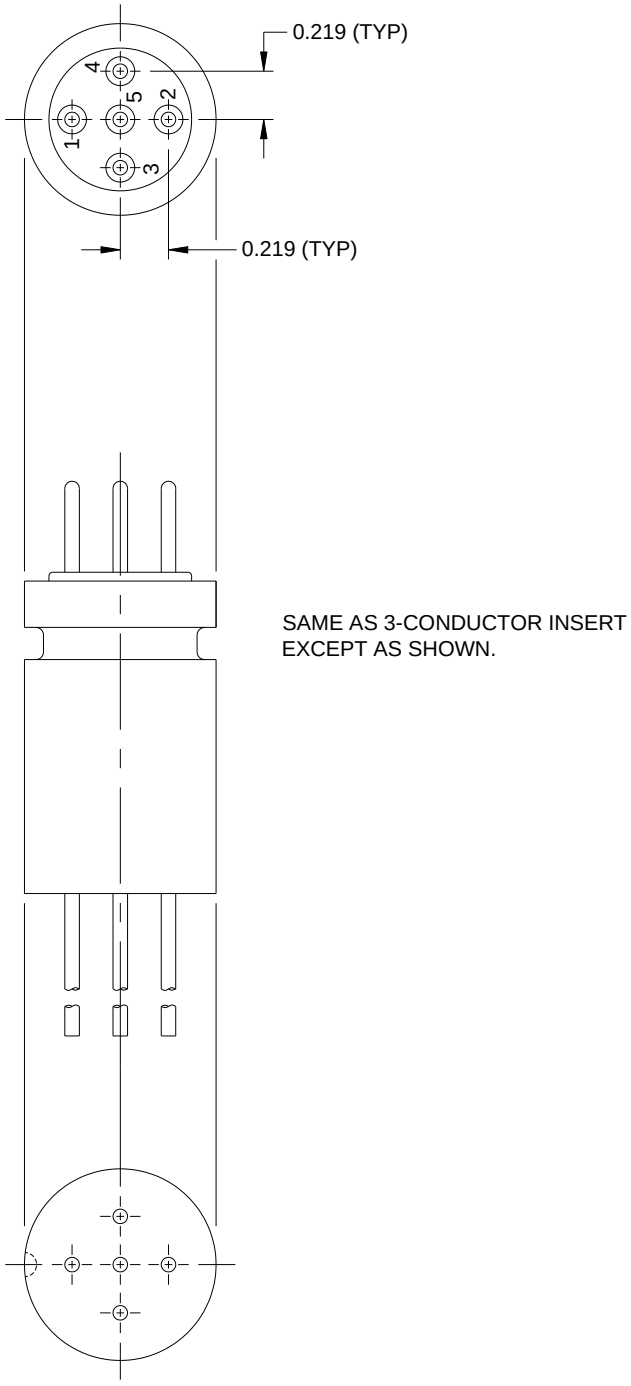


FIGURE 16. Assembly, insert molded (5-conductor) – M24231/5-067. (18)

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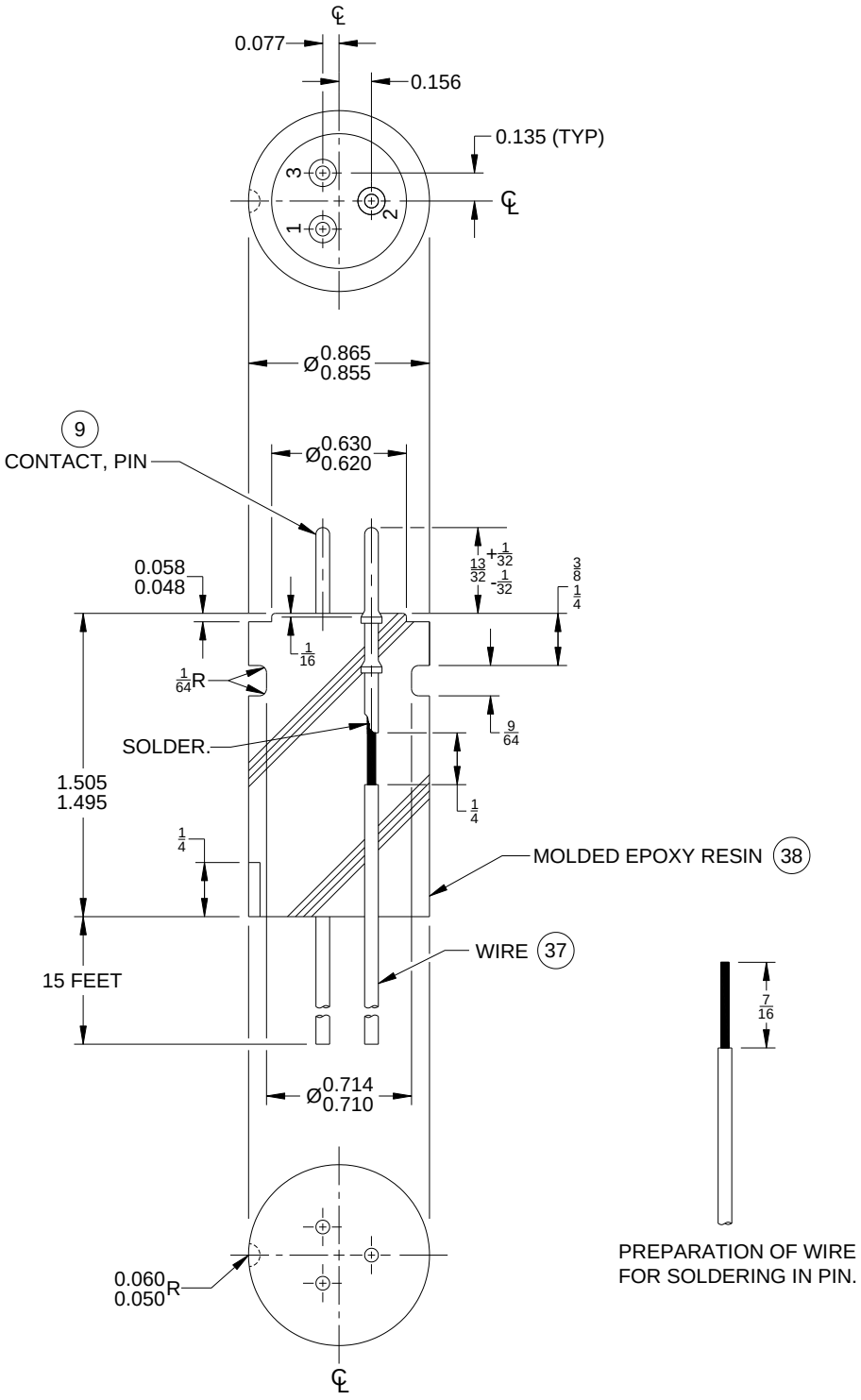


FIGURE 17. Assembly, insert, molded (3-conductor) – M24231/5-010. (16)

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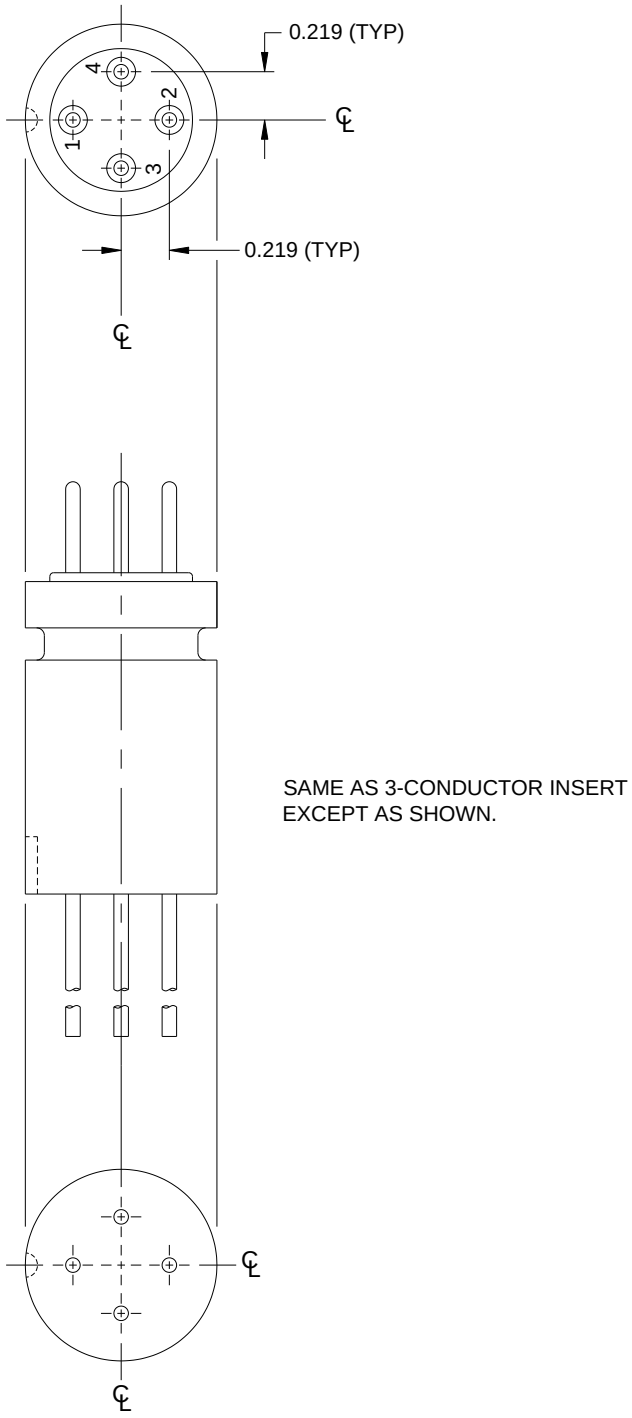


FIGURE 18. Assembly, insert, molded (4-conductor) – M24231/5-020. (17)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/5- 3/			Material/specification	Remarks
		001	002	003		
1	Connector assembly, single cable, 3-conductor, symbol no. 517	1			-----	See figure 1.
2	Connector assembly, single cable, 4-conductor, symbol no. 517.11		1		-----	See figure 1.
3	Connector assembly, single cable, 5-conductor			1	-----	See figure 1.
4	Gasket, O-ring	1	1	1	SAE-AS28775-215	Lubricate with silicone compound, SAE-AS8660; see figure 1.
5	Gasket, O-ring	1	1	1	SAE-AS28775-329	See item 4.
6	Gasket, O-ring	1	1	1	SAE-AS28775-115	See item 4.
7	Gasket, O-ring	1	1	1	SAE-AS28775-327	See item 4.
8	Body, single cable connector	1	1	1	Ni-Cu-Al, Class A, QQ-N-286	See figure 2 and continuation of figure 2.
9	Pin, contact	3	4	5	MIL-DTL-24231	See figure 17.
10	Assembly, plate, retainer	1			See items 26 and 33	See figure 6 and requirement 6.
11	Assembly, plate, retainer		1		See items 26 and 31	See figure 11 and requirement 6.
12	Assembly, plate, retainer			1	See items 26 and 32	See figure 15 and requirement 6.
13	Nut, gland	1	1	1	Ni-Al-BRZ, ASTM B150/B150M	See figure 8 and requirement 2.
14	Nut, body	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 7.
15	Washer, body, 3.5 od by 1.760 id by 0.162 thick	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 1, finish shall be 63 microinches.
16	Insert assembly, molded	1			See items 09, 37, and 38	See figure 17.
17	Insert assembly, molded		1		See items 09, 37, and 38	See figure 18.
18	Insert assembly, molded			1	See items 09, 37, and 38	See figure 16.
19	Insert, 0.125±0.002 diameter, lg to suit	1	1	1	Nylon, Group I, Class I, ASTM D4066	See figure 1.
20	Screw, 0.138-32 by 3/16 lg RHMS	2	2	2	CRES, coml	See figure 1.
21	Ring, retaining	1	1	1	SPIROLOX, PRD-99-M, or equal	See figure 1, M24231/5-073.

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TABLE I. Parts list – Continued.

Item no. [1/]	Description [2/]	Quantity for M24231/5- [3/]			Material/specification	Remarks
		001	002	003		
22	Ring, retaining	1	1	1	TRUARC, Type 5100-162-W, or equal	See figure 1, M24231/5-076.
23	Cap, pressure-proof	1	1	1	M24231/12-011	See figure 1.
24	Sleeving, 0.360 id by 0.020 thick	14’	14’	14’	Type F, Form U, Category 1, color clear, MIL-I-631	See figure 1.
25	Pin, guide, 0.092 diameter, lg to suit	1	1	1	Ni-Cu, Class A, QQ-N-281	See figure 1 and figure 3 and requirements 4 and 7.
26	Pin, positioning	1	1	1	Ni-Cu, Class A, QQ-N-281	See figure 6, figure 11, and figure 15.
27	Plate, identification, 1/16 thick	1	1	1	CRES or Al, Type 1, MIL-DTL-15024	See figure 12.
28	Plate, retainer	1			Ni-Cu, Class A, QQ-N-281	See figure 4 and requirement 6.
29	Plate, retainer		1		Ni-Cu, Class A, QQ-N-281	See figure 9 and requirement 6.
30	Plate, retainer			1	Ni-Cu, Class A, QQ-N-281	See figure 13 and requirement 6.
31	Plate, retainer		1		Ni-Cu, Class A, QQ-N-281	See figure 11 and requirement 6.
32	Plate, retainer			1	Ni-Cu, Class A, QQ-N-281	See figure 15 and requirement 6.
33	Plate, retainer	1			Ni-Cu, Class A, QQ-N-281	See figure 6 and requirement 6.
34	Packing, durometer 55±5	1			Rubber, ZZ-R-768	See figure 5.
35	Packing, durometer 55±5		1		Rubber, ZZ-R-768	See figure 10.
36	Packing, durometer 55±5			1	Rubber, ZZ-R-768	See figure 14.
37	Wire, 15’1”	3	4	5	MIL- DTL-16878/36 BJE0	See figure 17.
38	Epoxy resin, molded	As required			Cast epoxy resin comp [4/]	See figure 16, figure 17, and figure 18.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Arco Set 200, M.M. Reif and Co., Ivyland, PA, or PR610-TCF (Chockfast Orange), ITW Philadelphia Resins Corp., Montgomeryville, PA, or equal.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-054)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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